Skip-Aware LoRA Continuation for Pruned Diffusion Models

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Abstract

Text-to-image diffusion models typically require multiple denoising steps. Structured pruning and adaptive step skipping reduce the computational cost per step and the number of UNet evaluations, respectively. However, directly combining them can couple their errors along skipped trajectories and degrade perceptual consistency. We propose Skip-Aware LoRA Continuation, which builds on Stage1 LoRA repair and uses the full SDXL model as a teacher to perform latent-only trajectory matching on segments identified by Adaptive Skip. This lightweight continuation procedure adapts the latent evolution of the pruned model to skipped inference without modifying either the pruning algorithm or the skipping criterion. Experiments on SDXL using Prompt-100 and OBS pruning ratios from 10% to 60% show that Ours Stage2 + Skip consistently reduces LPIPS relative to direct OBS + Skip while maintaining comparable inference time and actual skip rate. The relative LPIPS reduction increases from 3.5% at OBS10 to 18.9% at OBS60, and PSNR improves at OBS20–OBS60. These results indicate that skip-aware continuation mitigates perceptual degradation along adaptively skipped trajectories, although it does not outperform conventional Stage1 LoRA repair on CLIPScore and DINO distance in every setting.

Keywords: Diffusion models, Structured pruning, Adaptive step skipping, LoRA, Efficient inference

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1. Introduction

Latent diffusion models, including Stable Diffusion and SDXL, have achieved substantial progress in open-domain text-to-image generation and are widely used for content creation, image editing, and design assistance. Nevertheless, generating an image typically requires dozens of iterative denoising steps. As image resolution and model size increase, repeated evaluations of the UNet denoiser incur substantial inference latency and GPU memory usage. Reducing inference costs while preserving generation quality is therefore an important problem in efficient generative modeling.

Existing acceleration approaches broadly fall into model compression and sampling acceleration. Model compression reduces per-step costs through pruning, quantization, distillation, or low-rank adaptation. Sampling acceleration reduces the number of UNet evaluations by improving the scheduler, decreasing the number of sampling steps, or dynamically skipping steps. These approaches appear naturally complementary: pruning reduces the cost of each step, while Adaptive Skip reduces the number of executed steps.

Our experimental observations suggest that this combination is not always stable. Pruning changes the denoiser's prediction errors across timesteps, whereas Adaptive Skip changes the latent trajectory. When a pruned model encounters a state distribution altered by step skipping, pruning-induced errors can be amplified during subsequent denoising, degrading texture, structure, or semantic detail. Thus, pruning and skipping are not entirely independent acceleration modules; they act on the same denoising trajectory.

To address this issue, we propose Skip-Aware LoRA Continuation. Rather than redesigning the pruning algorithm or the core Adaptive Skip criterion, we add a lightweight second training stage after conventional LoRA repair. Stage1 restores the pruned model's basic generation capability under conventional sampling. Stage2 explicitly simulates skipped segments and aligns the student's unrolled latent trajectory with that of the full SDXL teacher.

Our main contributions are summarized as follows:

- Observation and problem formulation: We identify interaction errors between structured pruning and adaptive step skipping along the denoising trajectory, beyond the independent contributions of the two mechanisms.
- Method: We propose Skip-Aware LoRA Continuation to adapt pruned diffusion models through skipped-segment trajectory matching without changing the pruning algorithm or the skipping criterion.
- Evaluation: We evaluate OBS pruning ratios from 10% to 60%, quality–speed trade-offs, threshold and LoRA-scale calibration, and qualitative examples. The perceptual benefits become more pronounced at moderate to high pruning ratios.

2. Related Work

2.1. Efficient Sampling for Diffusion Models

Diffusion models generate images through progressive denoising, usually requiring repeated evaluations of the denoising network. DDIM, DPM-Solver, and consistency-based sampling improve efficiency by modifying the sampling equations or reducing sampling steps. Adaptive Skip methods dynamically decide whether to skip denoising steps based on stability or prediction differences between neighboring timesteps, thereby reducing UNet evaluations. Such methods generally assume a sufficiently expressive and stable underlying denoiser. We investigate whether skipping strategies developed for full models remain suitable after structured pruning.

2.2. Pruning Diffusion Models

Model pruning reduces computation by removing less important parameters, channels, attention heads, or feed-forward neurons. Unlike classification networks, diffusion denoisers are repeatedly evaluated at different noise levels and timesteps. Pruning can therefore affect both overall representational capacity and timestep-dependent prediction errors. OBS-Diff uses second-order information to guide diffusion-model pruning, while EcoDiff and related work investigate

compression through learnable sparsity and post-pruning repair. Most existing studies evaluate pruned models along complete sampling trajectories and give less attention to their compatibility with additional adaptive step skipping.

2.3. LoRA Repair and Parameter-Efficient Fine-Tuning

LoRA enables parameter-efficient fine-tuning by introducing low-rank updates to pretrained weights. For an original weight matrix W, the update is expressed as

$$W' = W + \Delta W, \Delta W = BA \tag{1}$$

where A and B are trainable low-rank matrices. Since only a small number of additional parameters are optimized, LoRA is well suited to lightweight adaptation of large diffusion models. In pruned diffusion models, LoRA can restore basic generation capability. Conventional LoRA repair, however, does not explicitly account for trajectory changes introduced by adaptive skipping. Our second training stage addresses this gap.

3. Method

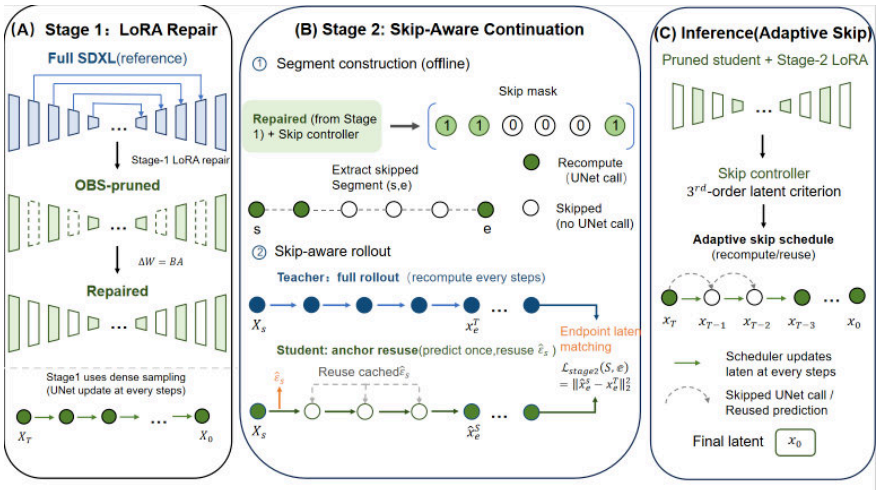


Figure 1: Overview of Skip-Aware LoRA Continuation: LoRA-based pruning repair, skip-aware continuation, and Adaptive Skip inference.

3.1. Problem Formulation

Let $\epsilon_\theta(x_t, t, c)$ denote the full diffusion denoiser, where x_t is the latent at timestep t and c is the text condition. The structurally pruned denoiser is denoted by $\epsilon_{\theta_p}(x_t, t, c)$. Conventional sampling progressively denoises along a sequence of timesteps. Adaptive Skip instead omits selected UNet evaluations according to a stability criterion and reuses the most recently cached noise prediction.

When pruning is directly combined with Adaptive Skip, the total error can be conceptually decomposed as

$$\Delta_{\text{total}} \approx \Delta_{\text{prune}} + \Delta_{\text{skip}} + \Delta_{\text{interaction}} \tag{2}$$

where Δ_{prune} denotes pruning-induced model error, Δ_{skip} denotes skipping-induced trajectory error, and $\Delta_{\text{interaction}}$ denotes their interaction when the pruned model follows a skipped trajectory. Our objective is to reduce $\Delta_{\text{interaction}}$ through lightweight LoRA continuation, without redesigning pruning or skipping rules.

3.2. Stage1: LoRA Repair

Stage1 applies LoRA repair to the pruned UNet. The backbone parameters remain frozen, and only the LoRA updates are trained. This stage restores basic generation capability under conventional sampling and brings the student closer to the full SDXL teacher. Importantly, Stage1 does not explicitly model Adaptive Skip and therefore does not guarantee stability along skipped trajectories.

3.3. Skipped-Segment Collection

The Adaptive Skip criterion is based on local stability of the latent trajectory during denoising. For each candidate timestep, we compute a relative third-order difference measure from neighboring latent states:

$$r_{i-1} = \frac{\|\Delta x_i\|_2 + \epsilon}{\|\Delta^{(3)} x_{i-1}\|_2}, \quad \Delta^{(3)} x_{i-1} = \Delta x_{i-1} - 2\Delta x_i + \Delta x_{i+1}, \quad \Delta x_i = x_i - x_{i+1}.$$

When $r_{i-1} < \tau$ and the current number of consecutive skipped steps remains below the limit K, the controller sets the binary mask to $m_{i-1} = 0$. The student UNet evaluation is omitted, and the scheduler updates the latent using the noise prediction cached at the most recent recomputation step. Otherwise, the student UNet is evaluated and $m_{i-1} = 1$. The skipping controller operates on the Stage1-repaired pruned student and does not require the full SDXL teacher. The teacher is used only to provide reference latent trajectories during Stage2 training. Final inference and main evaluation use $\tau_{\text{eval}} = 0.014$ and a maximum of $K = 4$ consecutive skipped steps. Segment collection for Stage2 uses $\tau_{\text{train}} = 0.010$.

Before Stage2 training, we run one Adaptive Skip sampling pass over all training prompts using the Stage1-repaired student and save the binary mask sequence m_i offline. A value of $m_i = 1$ indicates UNet recomputation, whereas $m_i = 0$ indicates a skipped evaluation that reuses the latest cached noise prediction.

We then extract segments (s, e) from the masks. The anchor step s satisfies $m_s = 1$, and the consecutive skipped interval $s + 1, \ldots, e - 1$ satisfies $m_i = 0$. The endpoint e is the next recomputation step or the trajectory boundary. The skip length is $l(s, e) = e - s - 1$, and only segments with $l(s, e) \geq 1$ are used for Stage2. The extracted segment set remains fixed throughout training and is not refreshed as LoRA parameters change.

3.4. Stage2: Skip-Aware LoRA Continuation

Stage2 initializes the student with the repaired LoRA parameters and continues training on trajectory segments that simulate skipped sampling. The full SDXL model serves as the teacher, and the pruned model with LoRA serves as the student. For a segment (s, e), the student computes a noise prediction only once, at the anchor step:

$$\hat{\epsilon}_s^S = \epsilon_{\theta_p+\phi}(x_s, t_s, c). \tag{3}$$

This anchor prediction is then reused throughout the segment to simulate the intermediate UNet evaluations omitted during Adaptive Skip inference:

$$\hat{x}^S_{k+1} = D(\hat{x}^S_k, t_k, \hat{\epsilon}^S_s), \quad k = s, \ldots, e-1. \tag{4}$$

The teacher recomputes the noise prediction at every timestep and provides a non-skipped reference trajectory:

$$\epsilon^T_k = \epsilon_\theta(x^T_k, t_k, c), \quad x^T_{k+1} = D(x^T_k, t_k, \epsilon^T_k). \tag{5}$$

Our final method uses a latent-only Stage2 loss that compares the latent states at the segment endpoint:

$$L_{\text{stage2}}(s, e) = \|\hat{x}^S_e - x^T_e\|_2^2. \tag{6}$$

Here, $\hat{x}^S_e$ is the student's endpoint latent obtained by reusing the anchor prediction, and x^T_e is the non-skipped reference latent obtained through teacher recomputation at the corresponding timesteps. Earlier experiments also considered anchor noise-prediction distillation,

$$L_\epsilon(s) = \|\hat{\epsilon}^S_s - \epsilon^T_s\|_2^2, \tag{7}$$

and the combined objective $L_{\text{latent}} + \lambda_\epsilon L_\epsilon$. The supplementary ablation indicates that adding an anchor noise-prediction term with $\lambda_\epsilon = 0.10$ does not consistently outperform latent-only training. We therefore discuss this term in the loss ablation rather than include it in the final method. During training, segments are randomly sampled from the collected set, and AdamW optimizes the LoRA parameters while all other parameters remain frozen.

Algorithm 1 Skip-Aware LoRA Continuation

```
Input: Full SDXL teacher epsilon_theta, pruned student epsilon_p, Stage1 LoRA Delta_phi1, prompts
Output: Stage2 LoRA Delta_phi2
1: initialize Delta_phi2 <- Delta_phi1
2: for each prompt x in X do
3: collect skipped segment (s, e) using Adaptive Skip(tau, K)
```

```
4: compute teacher trajectory from s to e with Full SDXL
5: compute student anchor prediction epsilon_hat_s_S with pruned UNet + LoRA
6: unroll student latent by reusing epsilon_hat_s_S over skipped steps
7: L_stage2 <- ||x_hat_e_S - x_e_T||_2^2
8: update LoRA parameters phi2 with AdamW
9: end for
10: return Delta_phi2
```

3.5. Threshold and LoRA-Scale Calibration

The Adaptive Skip threshold controls skipping aggressiveness. A higher threshold generally increases the actual skip rate and reduces inference time but can introduce larger trajectory errors. The LoRA scale determines the strength of the LoRA update during inference. Based on supplementary experiments, our main configuration uses $\tau_{\text{eval}} = 0.014$, a LoRA scale of 0.6, and a maximum of four consecutive skipped steps. Stage2 segment collection uses $\tau_{\text{train}} = 0.010$, while final inference and main evaluation use $\tau_{\text{eval}} = 0.014$. This configuration maintains an actual skip rate of approximately 48% across OBS10–OBS60, enabling perceptual consistency to be compared at similar inference efficiency.

4. Experimental Setup

Experiments use SDXL with OBS structured pruning at ratios of 10%, 20%, 30%, 40%, 50%, and 60%. The 50% and 60% settings probe the limits under heavier pruning. All SDXL evaluation images are generated at 1024×1024 resolution with 35 inference steps, a guidance scale of 5.0, and a fixed random seed of 42. Adaptive Skip allows at most four consecutive skipped steps. Reported timings include VAE decoding.

The main evaluation uses Prompt-100, a fixed set of 100 manually curated text prompts. It covers photography, product still life, natural landscapes, interiors, nighttime reflections, complex materials, animals and people, fantasy and science fiction, traditional painting, and stylized images. These categories

assess the effects of pruning and skipping on texture, geometry, illumination, reflections, semantic consistency, and style preservation.

Qualitative comparisons use Prompt-20, a fixed selection of 20 representative prompts from Prompt-100 determined before the experiments rather than selected after inspecting generated images. The main text presents the OBS30 qualitative grid; other pruning settings are included in the appendix.

Table 1: Experimental setup summary

Group	Parameter	Setting
Base	Model / scheduler	SDXL, DDIMScheduler, 35 steps, guidance scale 5.0, seed 42
	Resolution	1024 × 1024 for evaluation; Stage2 trained at 512 × 512
Compression	OBS pruning	Structured pruning on SDXL UNet; OBS10–OBS60
	Stage1 LoRA	EcoDiff-style repair, rank 8, 500 steps, lr 1 × 10^-5, scale 1.0
	Stage2 LoRA	Full SDXL teacher, 30 train prompts, 10 validation prompts, 500 steps
	Stage2 loss	Skip-aware unrolled latent matching only
Skip	Adaptive Skip	$\tau_{\text{train}} = 0.010$, $\tau_{\text{eval}} = 0.014$, max skip steps 4, final Stage2 scale 0.6
Evaluation	Prompt / metrics	Prompt-100 and Prompt-20; time, speedup, skip rate, CLIP, LPIPS, DINO, PSNR

We compare Full SDXL, OBS + Skip, Stage1 LoRA + Skip, and Ours Stage2 + Skip. OBS + Skip directly applies Adaptive Skip to the pruned model. Stage1 LoRA + Skip tests whether conventional pruning repair is sufficient. Ours Stage2 + Skip adds skip-aware latent-only continuation after Stage1 repair.

Evaluation metrics include average inference time, speedup relative to the full SDXL no-skip baseline, actual skip rate, CLIPScore, LPIPS to Full SDXL, DINO distance to Full SDXL, and PSNR to Full SDXL. Lower LPIPS and DINO distance are better; higher CLIPScore and PSNR are better. Larger speedup and skip rate indicate greater acceleration. We do not use FID as a primary metric because our objective is consistency with the full SDXL teacher under pruning and skipping, for which paired teacher-reference metrics are more directly relevant.

All experiments run on a single-GPU AutoDL server equipped with an NVIDIA GeForce RTX 4090 with 24 GB of memory. The driver version is 580.76.05, and nvidia-smi reports CUDA version 13.0. The Python environment uses Python 3.10.20, PyTorch 2.7.1+cu118, Torch CUDA 11.8, Diffusers 0.31.0,

Transformers 4.45.0, and Accelerate 1.12.0. xFormers is neither installed nor enabled; attention uses the default PyTorch/Diffusers implementation. Inference time is measured as per-image wall-clock time after warm-up and averaged over Prompt-100. It excludes model loading and includes the denoising loop and pipeline execution. OBS pruning, Stage1 repair, and Stage2 continuation are one-time offline costs and are excluded from inference speedup.

5. Results and Analysis

5.1. Main Results

Table 2: Main results across OBS pruning ratios. Time, speedup, and skip rate characterize inference efficiency. LPIPS, DINO distance, and PSNR use Full SDXL as the reference; CLIPScore measures text–image alignment.

Prune	Method	Time↓(s)	Speedup↑(%)	Skip Rate↑(%)	CLIP↑	LPIPS↓	DINO↓	PSNR↑
10%	OBS + Skip	2.74	42.56	48.50	0.3564	0.2728	0.1551	**21.64**
	Stage1 LoRA + Skip	2.71	43.16	48.84	0.3566	0.2767	0.1564	21.39
	Ours Stage2 + Skip	2.72	42.87	48.66	0.3550	**0.2632**	**0.1494**	21.62
20%	OBS + Skip	2.71	43.09	49.09	0.3550	0.3598	0.2255	19.54
	Stage1 LoRA + Skip	2.77	41.88	47.41	0.3583	0.3757	**0.2018**	19.00
	Ours Stage2 + Skip	2.73	42.71	48.38	0.3552	**0.3348**	0.2061	**19.73**
30%	OBS + Skip	2.76	42.06	47.84	0.3517	0.4168	0.3136	18.47
	Stage1 LoRA + Skip	2.71	43.15	48.84	0.3513	0.3956	0.2825	18.58
	Ours Stage2 + Skip	2.72	43.03	48.75	0.3516	**0.3802**	**0.2812**	**18.88**
40%	OBS + Skip	2.74	42.49	48.44	0.3438	0.4700	0.4017	18.06
	Stage1 LoRA + Skip	2.74	42.58	48.19	0.3485	0.4370	0.3512	18.07
	Ours Stage2 + Skip	2.73	42.68	48.31	0.3494	**0.4152**	**0.3406**	**18.33**
50%	OBS + Skip	2.76	42.15	48.25	0.3316	0.5262	0.5123	17.46
	Stage1 LoRA + Skip	2.72	42.97	48.88	0.3449	0.4555	**0.4022**	17.51
	Ours Stage2 + Skip	2.70	43.31	49.31	0.3407	**0.4458**	0.4349	**17.86**
60%	OBS + Skip	2.75	42.34	48.44	0.3033	0.5968	0.6784	16.76
	Stage1 LoRA + Skip	2.75	42.25	48.12	0.3285	0.5134	**0.5320**	16.96
	Ours Stage2 + Skip	2.77	41.99	47.81	0.3272	**0.4839**	0.5565	**17.50**

Table 3: LPIPS improvements of Ours Stage2 + Skip over OBS + Skip.

Prune	OBS + Skip LPIPS	Ours LPIPS	Relative improvement
10%	0.2728	0.2632	3.5%
20%	0.3598	0.3348	7.0%
30%	0.4168	0.3802	8.8%
40%	0.4700	0.4152	11.7%
50%	0.5262	0.4458	15.3%
60%	0.5968	0.4839	18.9%

Table 2 reports the complete main results for the three skipped-inference schemes across OBS10–OBS60. Ours Stage2 + Skip achieves inference times of approximately 2.70–2.77 s and actual skip rates of 47.8%–49.3%, comparable to OBS + Skip and Stage1 LoRA + Skip. Thus, the quality improvements are not obtained by reducing skipping or sacrificing inference speed.

The most consistent gains are observed in LPIPS. Relative to direct OBS + Skip, our method reduces LPIPS by 3.5%, 7.0%, 8.8%, 11.7%, 15.3%, and 18.9% at pruning ratios of 10%, 20%, 30%, 40%, 50%, and 60%, respectively. The relative improvement grows with pruning severity, suggesting that heavier pruning makes direct skipping more susceptible to perceptual trajectory degradation and increases the benefit of skip-aware latent adaptation.

PSNR largely supports this trend. At OBS10, our method is nearly tied with OBS + Skip, with a decrease of only 0.02 dB. At OBS20–OBS60, it consistently achieves higher PSNR; at OBS60, PSNR increases from 16.76 to 17.50 dB. DINO distance also decreases relative to OBS + Skip at every pruning ratio, indicating improved feature-space consistency.

Our method does not outperform Stage1 LoRA repair on every metric. At OBS50 and OBS60, for example, Stage1 achieves lower DINO distance, whereas our method achieves better LPIPS and PSNR. We therefore conclude that Stage2 improves perceptual consistency along skipped trajectories with almost no inference-speed penalty, consistently outperforming direct OBS + Skip on LPIPS and further improving Stage1 LPIPS and PSNR in most settings.

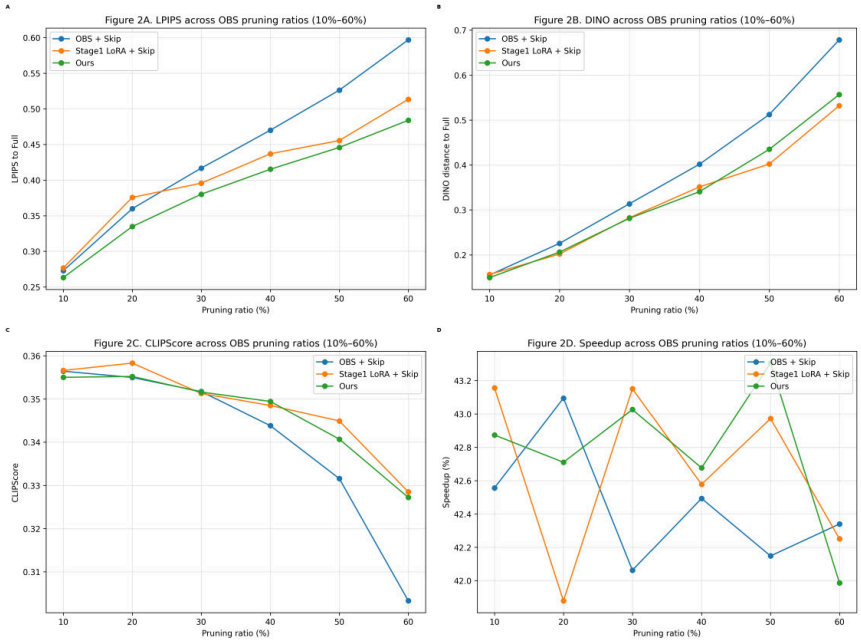


Figure 2: Quality–speed trends across pruning ratios. The benefits are more apparent at moderate to high pruning ratios.

5.2. LPIPS Trends and Quality–Speed Trade-Offs

Figure 3 compares LPIPS across the three skipped-inference schemes at different OBS pruning ratios. Ours Stage2 + Skip achieves lower LPIPS than direct OBS + Skip throughout OBS10–OBS60, with relative improvements increasing as pruning becomes more severe. The time, speedup, and skip-rate results in Table 2 show comparable inference efficiency across the three schemes, indicating that the LPIPS gains are not achieved by reducing skipping or sacrificing speed.

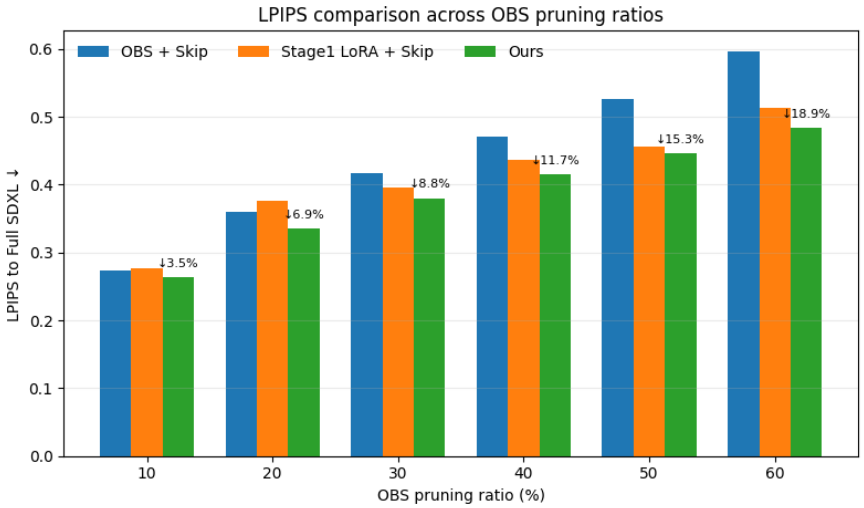


Figure 3: LPIPS comparison across OBS pruning ratios. Labels above the bars indicate the relative LPIPS improvement of Ours Stage2 + Skip over direct OBS + Skip.

5.3. Qualitative Comparisons

Figure 4 presents qualitative comparisons at OBS30. We use this setting in the main text because pruning damage is already visible while the model remains within a practically relevant moderate-to-high pruning regime. Examples cover curved structures, human scenes, interiors, complex textures, and high-frequency details. OBS30 + Skip is more susceptible to blur, structural drift, and local texture degradation. Stage1 LoRA + Skip recovers part of the baseline quality, while Ours Stage2 + Skip further improves detail stability at comparable inference speed and skip rate. The qualitative examples and representative Prompt-100 scenes suggest that our method is particularly useful for complex spatial layouts, rich materials, and strong illumination changes. Greenhouses, street markets, cathedrals, laboratories, snowy villages, and streets after rain combine multiple objects, transparent or reflective materials, haze, volumetric lighting, and high-frequency textures. Directly combining OBS pruning and Adaptive Skip can blur textures, distort local structures, and reduce lighting detail in these scenes. Latent trajectory adaptation on actual skipped segments

helps preserve local detail, material boundaries, and overall scene consistency.

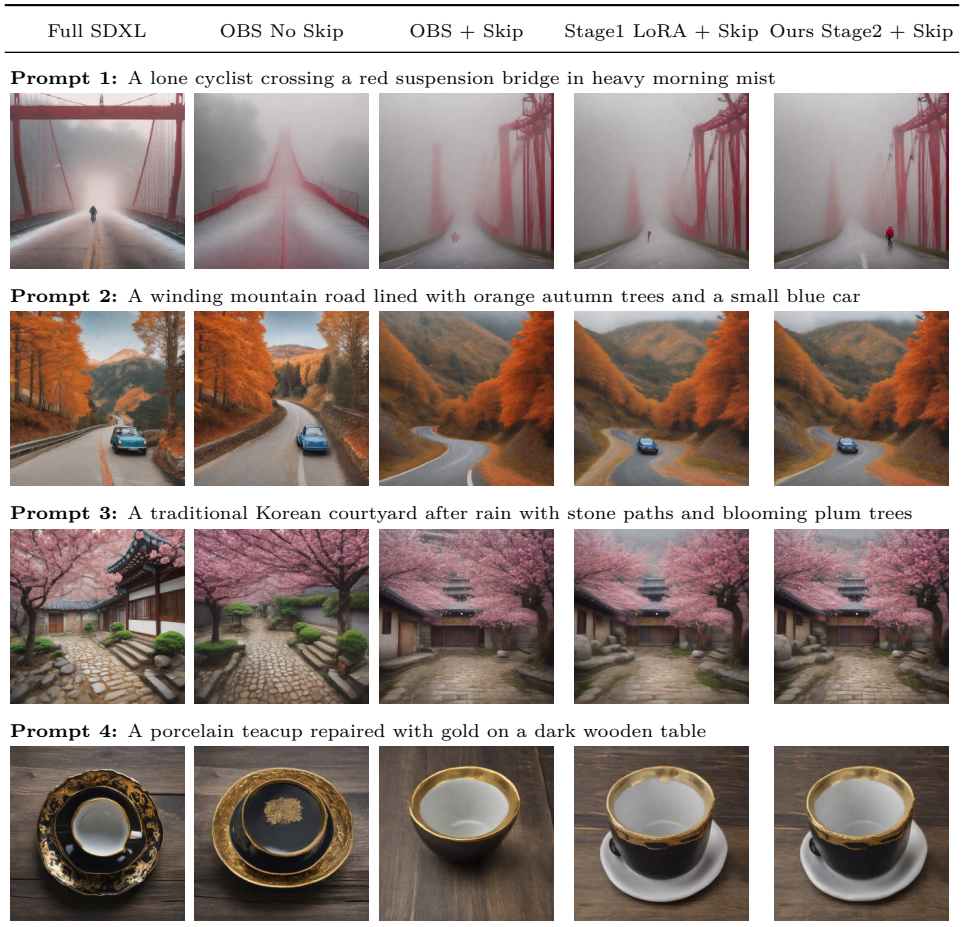


Figure 4: Qualitative comparisons at OBS30. Columns show Full SDXL, OBS No Skip, OBS30 + Adaptive Skip, Stage1 LoRA + Adaptive Skip, and Ours Stage2 + Adaptive Skip.

5.4. Ablation Studies

We perform ablations at OBS20 to examine the Stage2 objective and continuation strategy. All ablations use Prompt-100, $\tau_{\text{eval}} = 0.014$, a LoRA scale of 0.6, and a maximum of four consecutive skipped steps. Metrics follow the main evaluation: lower LPIPS and DINO distance and higher CLIPScore and PSNR are preferred, while larger speedup and actual skip rate indicate greater acceleration.

Loss ablation. Table 4 compares Stage2 loss designs. Anchor noise-prediction distillation alone achieves a lower DINO distance but worse LPIPS and PSNR than latent-only training. Compared with the noise-only objective, latent-only training reduces LPIPS from 0.3504 to 0.3348 and increases PSNR from 19.60 to 19.73 dB. Adding $0.10\mathcal{L}_\epsilon$ yields an LPIPS of 0.3448, which remains worse than latent-only training. Directly constraining endpoint latents therefore better serves our objective of mitigating skipped-trajectory drift, whereas the anchor noise-prediction term provides no consistent gain. We consequently use the latent-only objective in the final Stage2 method.

Table 4: Stage2 loss ablation at OBS20. Time is in seconds; speedup and skip rate are percentages.

Loss	Time↓	Speedup↑	Skip↑	CLIP↑	LPIPS↓	DINO↓	PSNR↑
$\mathcal{L}_\epsilon$ only	2.71	43.23	48.97	0.3547	0.3504	**0.2039**	19.60
$\mathcal{L}_{latent}$ only	2.73	42.73	48.38	**0.3552**	**0.3348**	0.2061	**19.73**
$\mathcal{L}_{latent} + 0.10\mathcal{L}_\epsilon$	2.73	42.77	48.44	0.3551	0.3448	0.2055	19.67

Continuation ablation. Table 5 compares continuation strategies. Stage1 only applies Adaptive Skip after conventional LoRA repair. Ordinary continuation adds training without explicitly distinguishing actual skipped segments. Skip-aware latent-only continuation adapts latent trajectories using segments produced by Adaptive Skip. Ordinary continuation improves LPIPS and PSNR over Stage1 only, but skip-aware continuation further reduces LPIPS to 0.3348 and raises PSNR to 19.73 dB. Relative to Stage1 only, it reduces LPIPS by 7.7% and improves PSNR by 0.35 dB; relative to ordinary continuation, it reduces LPIPS by 4.3% and improves PSNR by 0.11 dB. These results suggest that gains arise not solely from additional training but also from explicit adaptation to skipped trajectories. Stage1 only achieves a slightly higher CLIPScore, and ordinary continuation achieves a slightly lower DINO distance. Thus, skip-aware continuation primarily improves perceptual consistency and reconstruction along skipped trajectories rather than optimizing every metric simultaneously.

Table 5: Continuation-strategy ablation at OBS20. Time is in seconds; speedup and skip rate are percentages. Ours denotes skip-aware latent-only continuation.

Method	Time↓	Speedup↑	Skip↑	CLIP↑	LPIPS↓	DINO↓	PSNR↑
Stage1 only + Skip	2.74	42.54	48.16	**0.3584**	0.3625	0.2056	19.39
Ordinary cont + Skip	**2.72**	**42.91**	**48.59**	0.3547	0.3497	**0.2040**	19.63
Skip-aware + Skip	2.73	42.73	48.38	0.3552	**0.3348**	0.2061	**19.73**

Configuration Sensitivity. We analyze sensitivity to the Adaptive Skip threshold and LoRA scale at OBS20. This experiment uses 20 prompts to examine configuration trends; its absolute values are therefore not directly compared with the Prompt-100 main results. Table 6 reports LPIPS, PSNR, and inference efficiency for different thresholds and LoRA scales.

With the LoRA scale fixed at 0.6, increasing the threshold from 0.010 to 0.014 raises the actual skip rate from 42.50% to 46.56% and speedup from 37.30% to 40.63%. LPIPS changes only slightly, from 0.4018 to 0.4050, while PSNR remains nearly unchanged. Thus, $\tau = 0.014$ provides greater acceleration at similar reconstruction quality. With the threshold fixed at 0.014, increasing the LoRA scale from 0.6 to 0.8 increases LPIPS from 0.4050 to 0.4165 and decreases PSNR from 19.20 to 18.75 dB. We therefore use a threshold of 0.014 and a LoRA scale of 0.6 for the main experiments.

Table 6: Sensitivity to threshold and LoRA scale at OBS20, evaluated on 20 prompts to analyze configuration trends.

Setting	Time↓ (s)	Speedup↑ (%)	Skip↑ (%)	LPIPS↓	PSNR↑
Threshold sensitivity, scale = 0.6					
$\tau = 0.010$	2.99	37.30	42.50	**0.4018**	19.19
$\tau = 0.012$	2.89	39.51	45.00	0.4041	19.19
$\tau = 0.014$	**2.83**	**40.63**	**46.56**	0.4050	**19.20**
LoRA scale sensitivity, $\tau = 0.014$					
scale = 0.6	**2.83**	**40.63**	**46.56**	**0.4050**	**19.20**
scale = 0.7	2.87	39.84	45.63	0.4103	18.98
scale = 0.8	2.87	39.75	45.47	0.4165	18.75

6. Discussion

Our results indicate a coupling between pruning and adaptive step skipping. In a full model, the stability criterion used by Adaptive Skip relies on strong denoising capability. Reduced representational capacity after pruning can increase prediction errors at certain timesteps. When skipping changes the latent trajectory, pruning and skipping errors can jointly influence subsequent denoising.

Skip-Aware LoRA Continuation is lightweight and targeted. It neither retrains the full model nor changes pruning or skipping rules. Instead, it trains only LoRA parameters to adapt latent trajectories on the skipped segments encountered during inference. Its benefit grows with pruning severity: the relative LPIPS reduction over direct OBS + Skip increases from 3.5% at OBS10 to 18.9% at OBS60. This makes the approach particularly relevant when heavy pruning is combined with adaptive skipping.

Several limitations remain. First, our experiments primarily concern SDXL and OBS structured pruning. Although they cover pruning ratios from 10% to 60%, generalization to SD1.5, SD3.5, and other pruning methods requires further validation. Second, Stage2 introduces an additional offline training cost.

Although it does not affect deployed inference speed, this cost must be balanced against model size and application requirements. Third, generative-model evaluation is multidimensional: LPIPS, DINO distance, CLIPScore, and PSNR measure different properties. Stage2 primarily improves LPIPS and PSNR, while CLIPScore and DINO distance do not always surpass Stage1. The results should therefore be interpreted as improvements in skipped-trajectory and perceptual consistency, not universal superiority across metrics.

7. Conclusion

We investigate the compatibility of structured pruning and adaptive step skipping in diffusion models. Pruning, LoRA repair, and Adaptive Skip exhibit substantial interactions, and directly combining them does not always yield a favorable quality–speed trade-off. We propose a two-stage LoRA approach consisting of Stage1 pruning repair and Stage2 skip-aware continuation, which further aligns the pruned model with the full SDXL teacher along skipped trajectories. Results on Prompt-100 across OBS10–OBS60 show consistent LPIPS reductions at approximately 42% speedup and a 48% actual skip rate, with PSNR improvements at OBS20–OBS60. The relative LPIPS improvement over direct OBS + Skip increases from 3.5% at OBS10 to 18.9% at OBS60. These findings support continued training on skipped trajectories as a way to mitigate perceptual degradation in pruned diffusion models. Future work will explore additional pruning ratios, samplers, and diffusion backbones, together with human-preference and task-specific metrics to assess practical utility.

CRediT Authorship Contribution Statement

Xianwei Wu, Yupeng Wu, and Baoqian Tu: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review and editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data Availability

The prompt lists, evaluation scripts, and supplementary experimental materials are available upon reasonable request.

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Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT to assist with language polishing, translation, structural revision, and manuscript editing. After using this tool, the authors reviewed and revised the content as needed and take full responsibility for the content of the manuscript.

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Appendix A. Reproducibility Materials and Supplementary Results

Appendix A.1. Prompt Sets and Complete Lists

Prompt-100 is used for quantitative evaluation, and Prompt-20 is used for qualitative visualization. Both sets are fixed before experimentation rather than selected after inspecting generated results. Prompt-20 consists of the first 20 representative prompts in Prompt-100 and covers night scenes, still life, landscapes, interiors, materials, people, fantasy, science fiction, and traditional painting.

Table A.7: Summary of the prompt sets.

Prompt set	Number	Usage
Prompt-100	100	Quantitative metrics
Prompt-20	20	Qualitative grid

Complete Prompt-20 List

Table A.8: Complete Prompt-20 list.

No.	Prompt
1	A rain-soaked city avenue at night with neon reflections, umbrellas, and distant traffic lights
2	A photorealistic ceramic teapot on a wooden table beside fresh oranges and soft window light
3	A snowy mountain observatory with astronomers preparing a large telescope under a clear sky
4	A fantasy herbalist greenhouse filled with glowing plants, glass jars, and hand-written labels
5	A close-up macro photo of frost patterns spreading across a blue metal surface
6	A small fishing boat returning to a foggy harbor at sunrise with seagulls overhead
7	A futuristic hospital room with transparent displays, clean lighting, and medical robots
8	A medieval blacksmith forging a sword in a stone workshop full of sparks and smoke
9	A quiet Japanese garden after rain with mossy stones, koi pond, and red maple leaves
10	A high-end product photograph of a silver wristwatch on black slate with water droplets
11	A cyberpunk noodle shop in a narrow alley with steam, lanterns, and glowing signs
12	A dramatic oil painting of a queen in emerald armor standing in a ruined courtyard
13	A documentary photo of a chef plating colorful desserts in a bright restaurant kitchen
14	A lonely lighthouse on dark volcanic cliffs under storm clouds and crashing waves
15	A traditional Chinese ink painting of cranes flying above layered misty mountains
16	A luxury spacecraft cockpit overlooking Saturn's rings with cool interior lighting
17	A handmade wooden toy train on a child's bedroom floor in warm afternoon light
18	A surreal underwater ballroom with chandeliers, drifting curtains, and silver fish
19	A renaissance portrait of an old scientist holding a glowing crystal orb
20	A winding mountain road lined with orange autumn trees and a small blue car

Complete Prompt-100 List

Table A.9: Complete Prompt-100 list.

No.	Prompt
1	A rain-soaked city avenue at night with neon reflections, umbrellas, and distant traffic lights
2	A photorealistic ceramic teapot on a wooden table beside fresh oranges and soft window light
3	A snowy mountain observatory with astronomers preparing a large telescope under a clear sky
4	A fantasy herbalist greenhouse filled with glowing plants, glass jars, and hand-written labels
5	A close-up macro photo of frost patterns spreading across a blue metal surface
6	A small fishing boat returning to a foggy harbor at sunrise with seagulls overhead

7 A futuristic hospital room with transparent displays, clean lighting, and medical robots
8 A medieval blacksmith forging a sword in a stone workshop full of sparks and smoke
9 A quiet Japanese garden after rain with mossy stones, koi pond, and red maple leaves
10 A high-end product photograph of a silver wristwatch on black slate with water droplets
11 A cyberpunk noodle shop in a narrow alley with steam, lanterns, and glowing signs
12 A dramatic oil painting of a queen in emerald armor standing in a ruined courtyard
13 A documentary photo of a chef plating colorful desserts in a bright restaurant kitchen
14 A lonely lighthouse on dark volcanic cliffs under storm clouds and crashing waves
15 A traditional Chinese ink painting of cranes flying above layered misty mountains
16 A luxury spacecraft cockpit overlooking Saturn's rings with cool interior lighting
17 A handmade wooden toy train on a child's bedroom floor in warm afternoon light
18 A surreal underwater ballroom with chandeliers, drifting curtains, and silver fish
19 A renaissance portrait of an old scientist holding a glowing crystal orb
20 A winding mountain road lined with orange autumn trees and a small blue car
21 A highly detailed pocket watch opened on velvet with visible gears and engraved metal
22 A peaceful tea house beside a bamboo forest during early morning fog
23 A neon arcade room filled with polished floors, colorful machines, and reflective lights
24 A realistic tabby cat sitting on a piano surrounded by scattered rose petals
25 A deep jungle temple entrance covered in vines and shafts of golden light
26 A modern apartment living room with floor-to-ceiling windows and morning sunlight
27 A bowl of ramen on a wooden counter with steam, chopsticks, and soft restaurant lighting
28 A mountain village covered in fresh snow beneath a starry winter sky
29 A tiny dragon sleeping on ancient books in a candlelit library
30 A young astronaut repairing a satellite while Earth glows in the background
31 A futuristic tram station at night with wet pavement and blue neon signs
32 A medieval alchemist workshop with glowing bottles, notes, and detailed shelves
33 A golden retriever wearing aviator goggles inside an old airplane hangar
34 A luminous jellyfish drifting through a dark underwater cave
35 A luxury perfume bottle on a reflective surface with soft studio lighting
36 A cozy attic art studio with canvases, plants, skylights, and warm light
37 A grand cathedral interior with stained glass windows and dust in sunbeams
38 A paper origami bird flying through a forest made of folded colorful paper
39 A desert archaeological camp beside ancient ruins during a red sunset
40 A sleek futuristic sports car driving across an elevated highway above a megacity
41 A cute hamster wearing a tiny wizard hat and holding a glowing wand
42 A complex glass butterfly sculpture shattering into colorful crystals
43 A peaceful river through an autumn forest with red leaves floating on the water
44 A bustling alien street market with neon food stalls and strange architecture
45 A pencil sketch style image of a futuristic greenhouse skyscraper
46 A mythical phoenix made of fire rising above golden ash
47 A wooden cabin near a snowy fjord with smoke from the chimney
48 A macro close-up of brass gears inside a steampunk clock
49 A floating island with waterfalls falling into clouds below
50 A vintage detective office with foggy windows, a typewriter, and low-key lighting
51 A polar bear on a small ice sheet under green northern lights
52 A glass-roofed botanical station on Mars filled with orange dust and tropical plants

53 A ceramic artist shaping a large blue vase in a quiet studio
54 A cable car passing above a snowy mountain town at golden hour
55 A cybernetic violinist performing alone on an empty subway platform
56 Rain droplets on a red maple leaf floating in a stone fountain
57 A desert observatory with brass telescopes under a purple twilight sky
58 A kitchen table covered with handmade dumplings, flour, and warm side light
59 A fantasy courier riding a white stag across a frozen lake beneath pale moons
60 A minimalist bedroom with linen bedding, paper screens, and morning shadows
61 A weathered fishing boat tied to a wooden pier in dense coastal fog
62 A tiny bookstore hidden inside an old train carriage during a rainy evening
63 A firefighter helmet resting on wet pavement after a storm
64 A floating market on a tropical river with lanterns, fruit boats, and misty hills
65 A marble statue workshop with unfinished sculptures and dust in sunbeams
66 A futuristic laboratory with transparent screens and soft blue surgical lighting
67 A quiet greenhouse cafe with hanging ferns, wicker chairs, and stained glass
68 A surreal clock tower melting into a field of yellow wildflowers
69 A silver fountain pen beside handwritten letters on cream paper
70 A deep-sea research base window looking out at glowing coral and dark water
71 A medieval bakery at dawn with baskets of bread and smoke from ovens
72 A lone cyclist crossing a red suspension bridge in heavy morning mist
73 A baroque theater stage with velvet curtains, candles, and an empty orchestra pit
74 A small orange robot repairing solar panels on the roof of a desert house
75 A traditional Korean courtyard after rain with stone paths and blooming plum trees
76 Aerial view of terraced rice fields reflecting a cloudy sunset
77 A porcelain teacup repaired with gold on a dark wooden table
78 A polar research tent glowing warmly under aurora lights
79 A still life of peaches, glassware, and folded silk in renaissance lighting
80 A sunlit pottery market with colorful bowls stacked on wooden tables
81 A futuristic rescue drone hovering above a flooded city street
82 A quiet monk walking through a bamboo grove after rainfall
83 A portrait of a woman wearing mirrored ski goggles in a snowstorm
84 A Victorian greenhouse at night illuminated by hundreds of small lamps
85 A handmade chess set carved from dark stone and ivory on velvet fabric
86 A spaceship repair bay filled with cables, sparks, and orange warning lights
87 A lonely desert highway motel under a huge Milky Way sky
88 A watercolor scene of sailboats resting in a harbor at low tide
89 A fantasy shop crowded with roots, glass jars, and glowing labels
90 A small mountain chapel surrounded by autumn trees and drifting fog
91 A chef tossing noodles in a steel wok with flames rising around the pan
92 A fox standing in tall grass beside an abandoned railway track
93 A crystal cave with a narrow wooden bridge and blue light from below
94 A retro computer room with beige monitors and sunlight through blinds
95 A detailed silver dragonfly brooch pinned to dark green velvet
96 A stormy ocean wave crashing against a red lighthouse at dusk
97 A cozy winter reading corner with a wool blanket and snow outside the window
98 A museum hall with dinosaur skeletons and moonlight from skylights

99	A low-angle photograph of a rain-covered motorcycle parked beside a brick wall
100	A fantasy mountain gate carved into black stone with blue torches

Appendix A.2. Supplementary Material Overview

Table A.10: Overview of supplementary materials.

Item	Description
Prompt sets	Prompt-100 for quantitative evaluation and Prompt-20 for qualitative visualization.
OBS20 metrics	Complete metrics for the main OBS20 configuration to reproduce the result tables.
Pruning-ratio summary	Complete metrics across OBS10–OBS60.
Supplementary grids	Additional qualitative comparisons on SDXL, SD1.5, and SD2.1.

Appendix A.3. Main Experiment Reproduction Workflow

Table A.11: Overview of the main experiment reproduction workflow.

Step	Key setting
Stage1 LoRA repair	Repair the pruned UNet with rank-8 LoRA for 500 steps at a learning rate of 1×10^{-5}.
Stage2 continuation	Initialize from Stage1 LoRA, collect skipped segments using $\tau_{\text{train}} = 0.010$, and train for 500 steps with the latent-only loss.
Adaptive Skip evaluation	Use $\tau_{\text{eval}} = 0.014$, at most four consecutive skipped steps, and a LoRA scale of 0.6; report results on Prompt-100.
Metric computation	Compute CLIPScore for text–image alignment and LPIPS, DINO distance, and PSNR using Full SDXL as the paired teacher reference.

The threshold $\tau_{\text{train}} = 0.010$ is used for skipped-segment collection during Stage2 training, whereas $\tau_{\text{eval}} = 0.014$ is used for final inference evaluation. These thresholds belong to different stages and should not be interpreted as conflicting settings for the same stage.

Appendix A.4. Complete OBS20 Results

Table A.12: Complete metrics at OBS20.

Method	Time	Speedup	Skip	CLIP↑	LPIPS↓	DINO↓	PSNR↑
OBS20 No Skip	4.70	–	0.00	0.3576	0.3439	0.2028	19.47
OBS20 + Skip	2.71	43.09	49.09	0.3550	0.3598	0.2255	19.54
Stage1 LoRA + Skip	2.77	41.88	47.41	0.3583	0.3757	0.2018	19.00
Ours Stage2 + Skip	2.73	42.71	48.38	0.3552	0.3348	0.2061	19.73

Appendix A.5. Threshold and LoRA-Scale Settings

The final main results use $\tau_{\text{eval}} = 0.014$, a LoRA scale of 0.6, and a maximum of four consecutive skipped steps. Segment collection for Stage2 uses $\tau_{\text{train}} = 0.010$, while final inference and evaluation use $\tau_{\text{eval}} = 0.014$. The full threshold/scale calibration table in an earlier manuscript version used an earlier experimental configuration and is not used as evidence for the present main results. This distinction avoids mixing that configuration with the OBS10–OBS60 latent-only results.

Appendix A.6. Additional Ablations of Ordinary and Skip-Aware Continuation

To test whether Stage2 gains arise solely from additional training, we introduce an ordinary-continuation baseline. It uses the same Stage1 LoRA initialization and the same 500 additional training steps as skip-aware latent-only continuation. The strategies differ in their objectives and trajectory construction. Ordinary continuation uses anchor noise-prediction distillation without explicitly constructing skipped trajectories; skip-aware latent-only continuation simulates segments produced by Adaptive Skip and matches endpoint latents. This comparison separates the effect of additional training from that of skipped-trajectory adaptation.

Table A.13: Additional continuation ablation at OBS30.

Method	Time	Skip	CLIP↑	LPIPS↓	DINO↓	PSNR↑
Stage1 only	2.720	49.06	0.3505	0.3952	0.2945	18.749
Ordinary continuation	2.731	48.66	0.3514	0.3834	**0.2679**	18.817
Skip-aware latent-only	2.729	48.75	**0.3516**	**0.3802**	0.2812	**18.876**

Table A.14: Additional continuation ablation at OBS40.

Method	Time	Skip	CLIP↑	LPIPS↓	DINO↓	PSNR↑
Stage1 only	2.744	48.50	0.3477	0.4348	0.3620	18.312
Ordinary continuation	**2.731**	**48.72**	**0.3503**	0.4240	**0.3396**	18.279
Skip-aware latent-only	2.751	48.31	0.3494	**0.4152**	0.3406	**18.333**

Additional training is beneficial: ordinary continuation reduces LPIPS and DINO distance relative to Stage1 only. Nevertheless, with the same initialization and additional training budget, skip-aware latent-only continuation further reduces LPIPS and improves PSNR at OBS30 and OBS40. The gains therefore cannot be explained entirely by longer training and are associated with explicit adaptation to skipped denoising trajectories. DINO distance and CLIPScore are not consistently best for the skip-aware method, indicating that Stage2 primarily improves perceptual consistency and pixel/trajectory recovery rather than all semantic-feature metrics.

Appendix A.7. Complete Metrics across Pruning Ratios

Table A.15: Complete metrics across OBS10–OBS60.

Prune	Method	Skip?	Time	Speedup	Skip	CLIP↑	LPIPS↓	DINO↓	PSNR↑
10%	No Skip	No	4.70	–	0.00	0.3581	0.2535	0.1480	21.65
10%	OBS + Skip	Yes	2.74	42.56	48.50	0.3564	0.2728	0.1551	21.64
10%	Stage1 LoRA + Skip	Yes	2.71	43.16	48.84	0.3566	0.2767	0.1564	21.39
10%	Ours Stage2 + Skip	Yes	2.72	42.87	48.66	0.3550	0.2632	0.1494	21.62
20%	No Skip	No	4.70	–	0.00	0.3576	0.3439	0.2028	19.47
20%	OBS + Skip	Yes	2.71	43.09	49.09	0.3550	0.3598	0.2255	19.54
20%	Stage1 LoRA + Skip	Yes	2.77	41.88	47.41	0.3583	0.3757	0.2018	19.00
20%	Ours Stage2 + Skip	Yes	2.73	42.71	48.38	0.3552	0.3348	0.2061	19.73
30%	No Skip	No	4.70	–	0.00	0.3542	0.4023	0.2863	18.40
30%	OBS + Skip	Yes	2.76	42.06	47.84	0.3517	0.4168	0.3136	18.47
30%	Stage1 LoRA + Skip	Yes	2.71	43.15	48.84	0.3513	0.3956	0.2825	18.58
30%	Ours Stage2 + Skip	Yes	2.72	43.03	48.75	0.3516	0.3802	0.2812	18.88
40%	No Skip	No	4.70	–	0.00	0.3486	0.4541	0.3748	18.05
40%	OBS + Skip	Yes	2.74	42.49	48.44	0.3438	0.4700	0.4017	18.06
40%	Stage1 LoRA + Skip	Yes	2.74	42.58	48.19	0.3485	0.4370	0.3512	18.07
40%	Ours Stage2 + Skip	Yes	2.73	42.68	48.31	0.3494	0.4152	0.3406	18.33
50%	No Skip	No	4.70	–	0.00	0.3372	0.5126	0.4921	17.47
50%	OBS + Skip	Yes	2.76	42.15	48.25	0.3316	0.5262	0.5123	17.46
50%	Stage1 LoRA + Skip	Yes	2.72	42.97	48.88	0.3449	0.4555	0.4022	17.51
50%	Ours Stage2 + Skip	Yes	2.70	43.31	49.31	0.3407	0.4458	0.4349	17.86
60%	No Skip	No	4.70	–	0.00	0.3101	0.5879	0.6548	16.78
60%	OBS + Skip	Yes	2.75	42.34	48.44	0.3033	0.5968	0.6784	16.76
60%	Stage1 LoRA + Skip	Yes	2.75	42.25	48.12	0.3285	0.5134	0.5320	16.96
60%	Ours Stage2 + Skip	Yes	2.77	41.99	47.81	0.3272	0.4839	0.5565	17.50

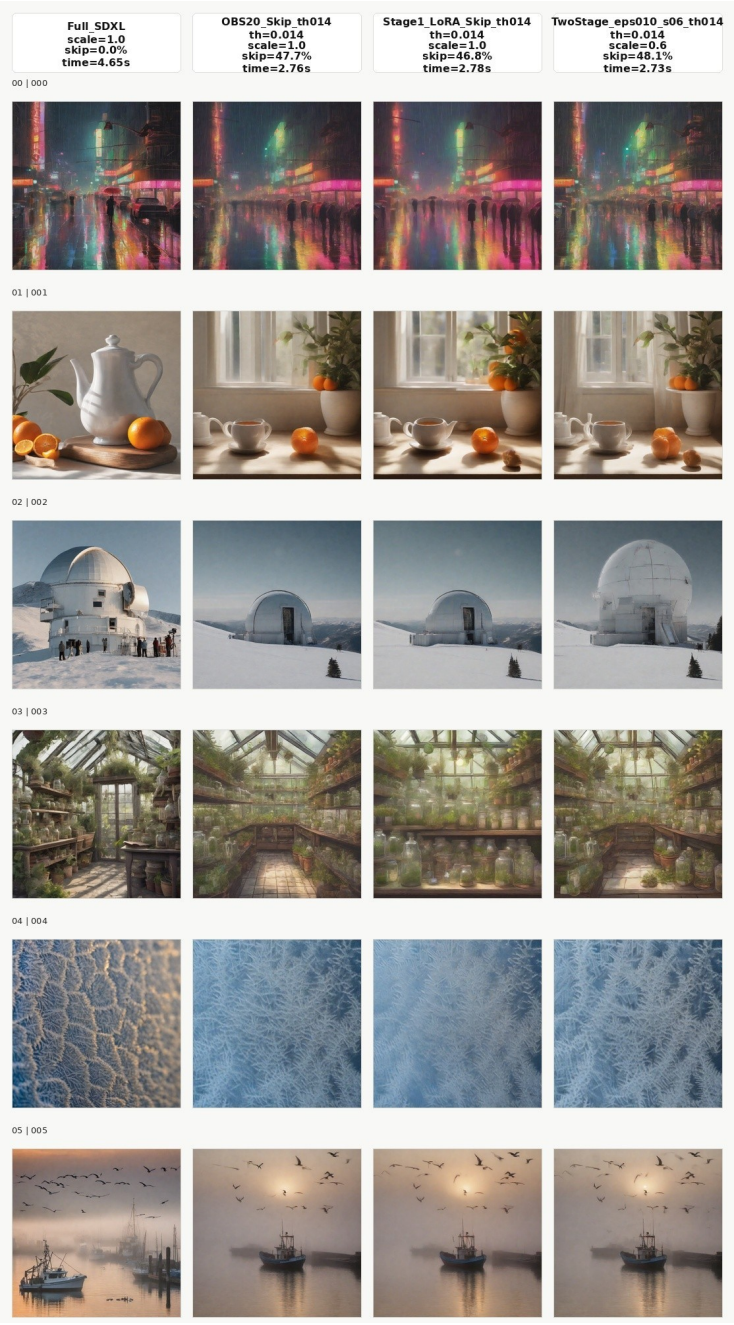


Figure A.5: Qualitative comparison of the main methods at OBS20.

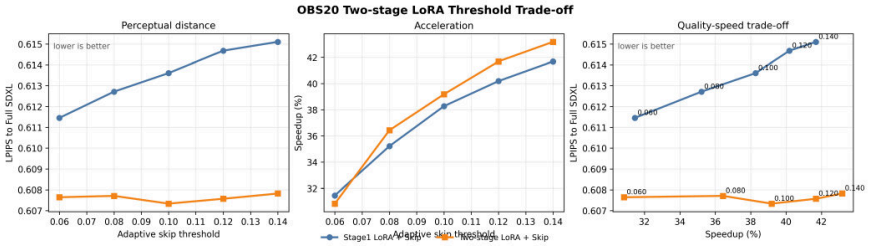


Figure A.6: Quality–speed trade-offs across thresholds at OBS20.

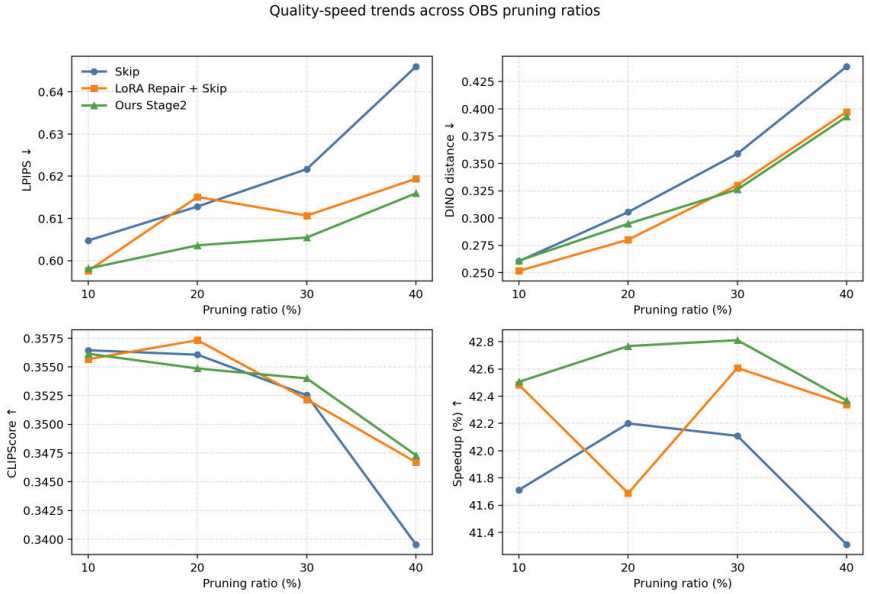


Figure A.7: Overview of quality–speed trade-offs across pruning ratios.

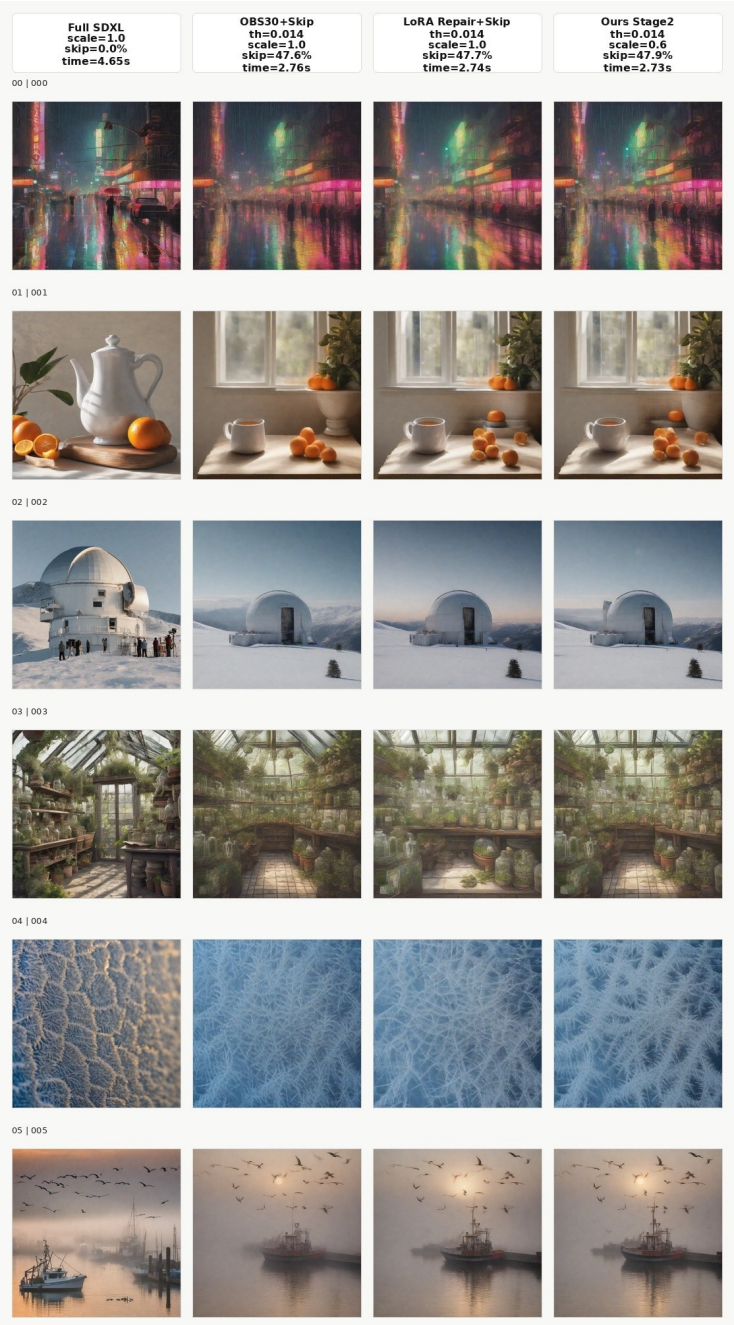


Figure A.8: Additional qualitative comparisons at OBS30.

Appendix A.9. Cross-Backbone Qualitative Results on SD1.5 and SD2.1

We provide supplementary qualitative results on SD2.1 and SD1.5 to preliminarily assess applicability across Stable Diffusion backbones. These examples illustrate the effects of skip-aware continuation on local structure, texture, and lighting stability at different pruning ratios. The main quantitative analysis remains based on the SDXL experiments.

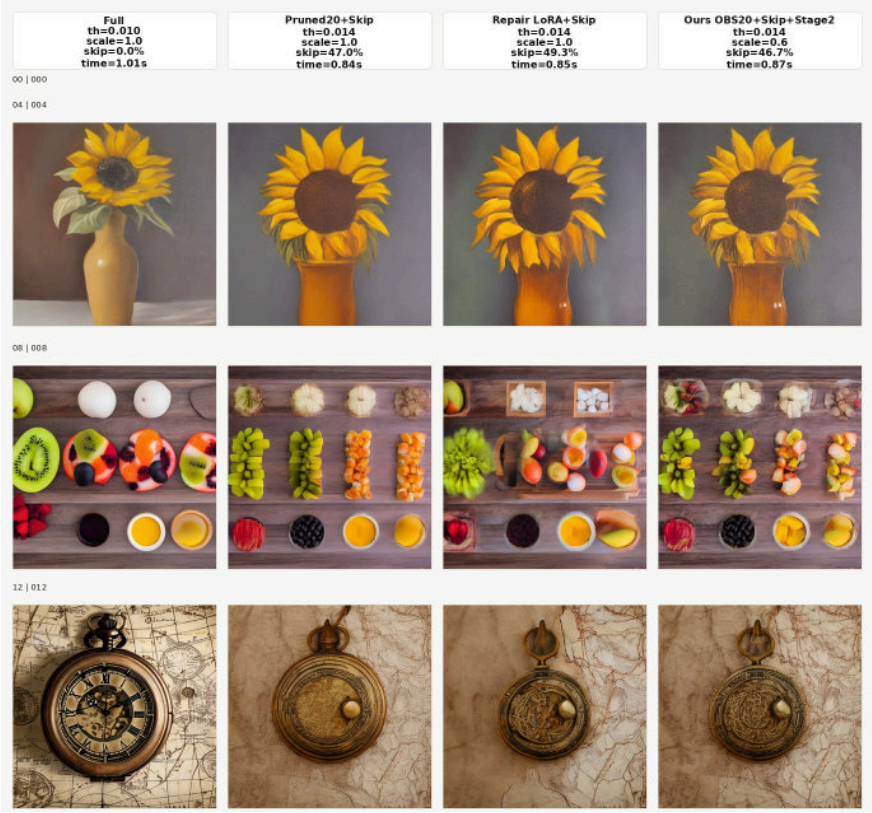


Figure A.9: Additional qualitative comparisons on SD2.1 at OBS20. Columns show Full, Pruned + Skip, Repair LoRA + Skip, and Ours Stage2 + Skip.

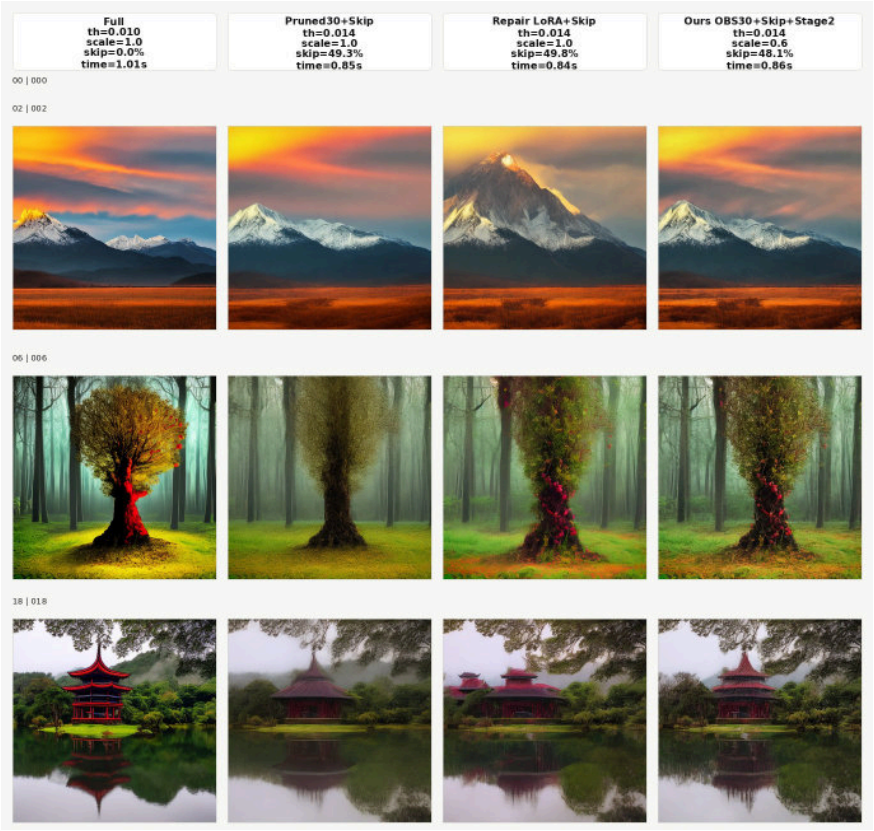


Figure A.10: Additional qualitative comparisons on SD2.1 at OBS30. Columns show Full, Pruned + Skip, Repair LoRA + Skip, and Ours Stage2 + Skip.

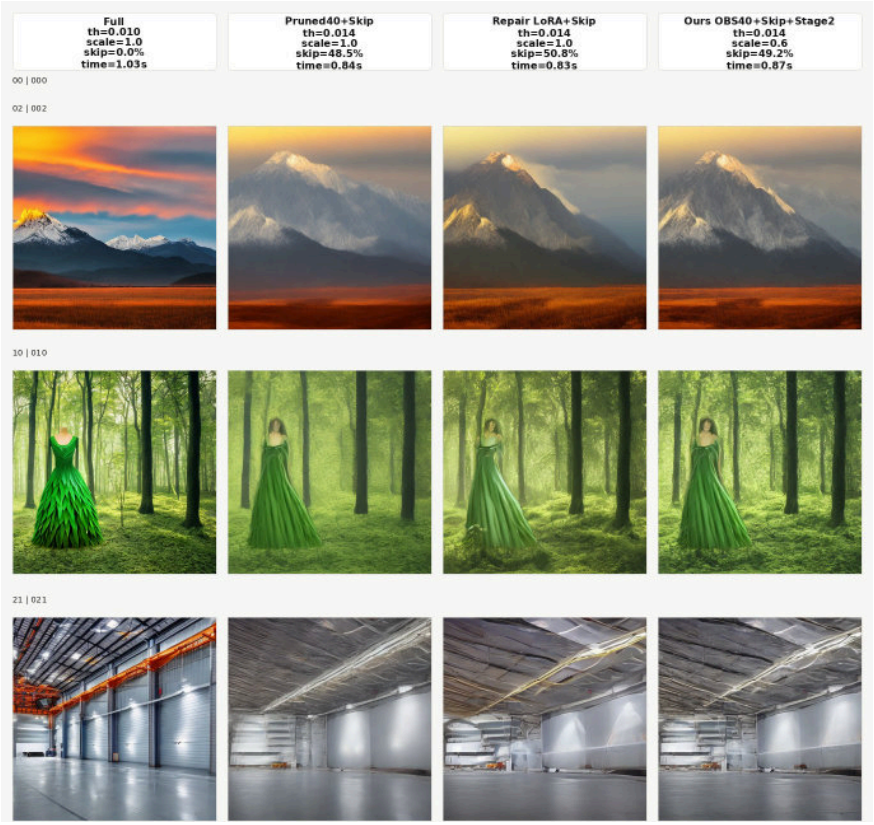


Figure A.11: Additional qualitative comparisons on SD2.1 at OBS40. Columns show Full, Pruned + Skip, Repair LoRA + Skip, and Ours Stage2 + Skip.

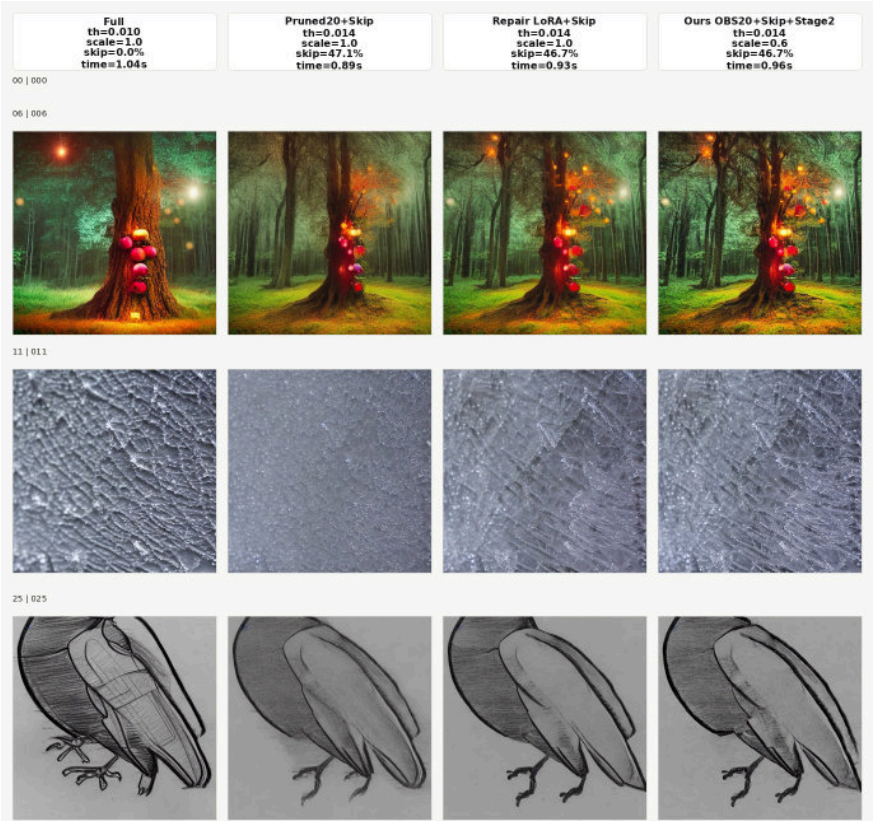


Figure A.12: Additional qualitative comparisons on SD1.5 at OBS20. Columns show Full, Pruned + Skip, Repair LoRA + Skip, and Ours Stage2 + Skip.

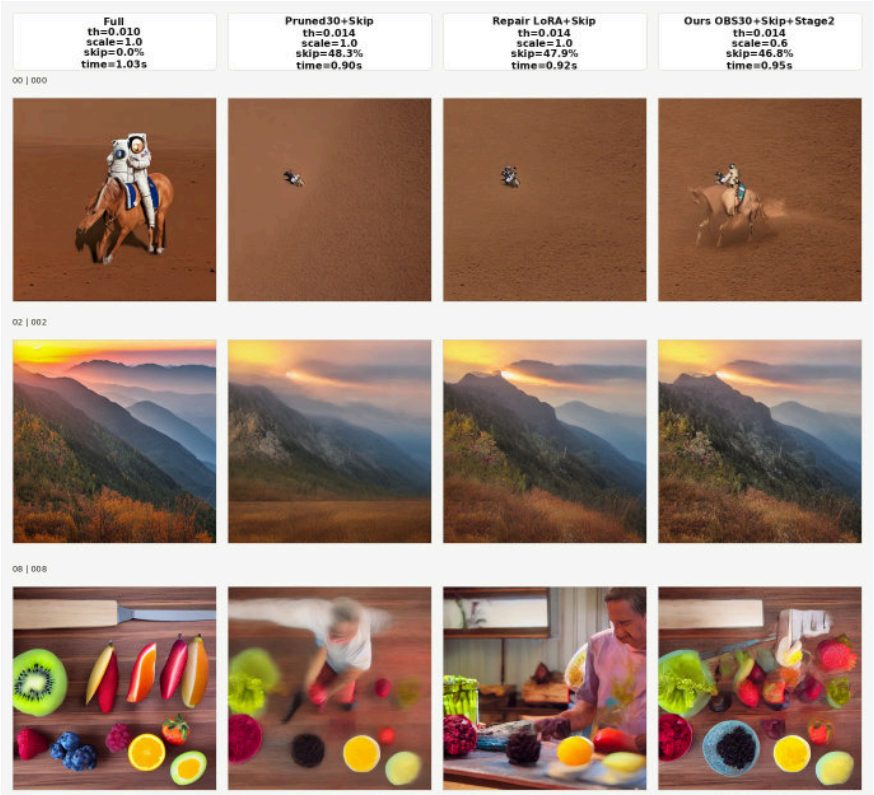


Figure A.13: Additional qualitative comparisons on SD1.5 at OBS30. Columns show Full, Pruned + Skip, Repair LoRA + Skip, and Ours Stage2 + Skip.



Figure A.14: Additional qualitative comparisons on SD1.5 at OBS40. Columns show Full, Pruned + Skip, Repair LoRA + Skip, and Ours Stage2 + Skip.